

Retail / E-commerce Performance Analysis

Sales, Profitability, and Discount Impact (2014–2017)

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Business Problem

- **Sales performance** has been growing over time, but profitability does not always follow the same trend.
- **Discounting** is widely used across product categories, but its impact on profit margins is unclear.
- Different **product categories** may perform differently in terms of revenue and profitability.
- The business needs to understand how **sales, profit, and discounting** interact to make better decisions.

Dataset Overview & Structure

Dataset: Superstore retail dataset (2014–2017)

- Contains transactional sales data across multiple product categories
- Each row represents a single customer order
- Used to analyse sales performance, profitability, and discounting behaviour

*Note: Analysis focuses on revenue and profit metrics; marketing cost data is not included.

Key Fields:

- Sales – total revenue per order
- Profit – profit per order after costs
- Profit Margin – profit as a percentage of sales
- Discount – percentage reduction applied to the order
- Category – product category (Furniture, Office Supplies, Technology)
- Order Date – used for time-based analysis

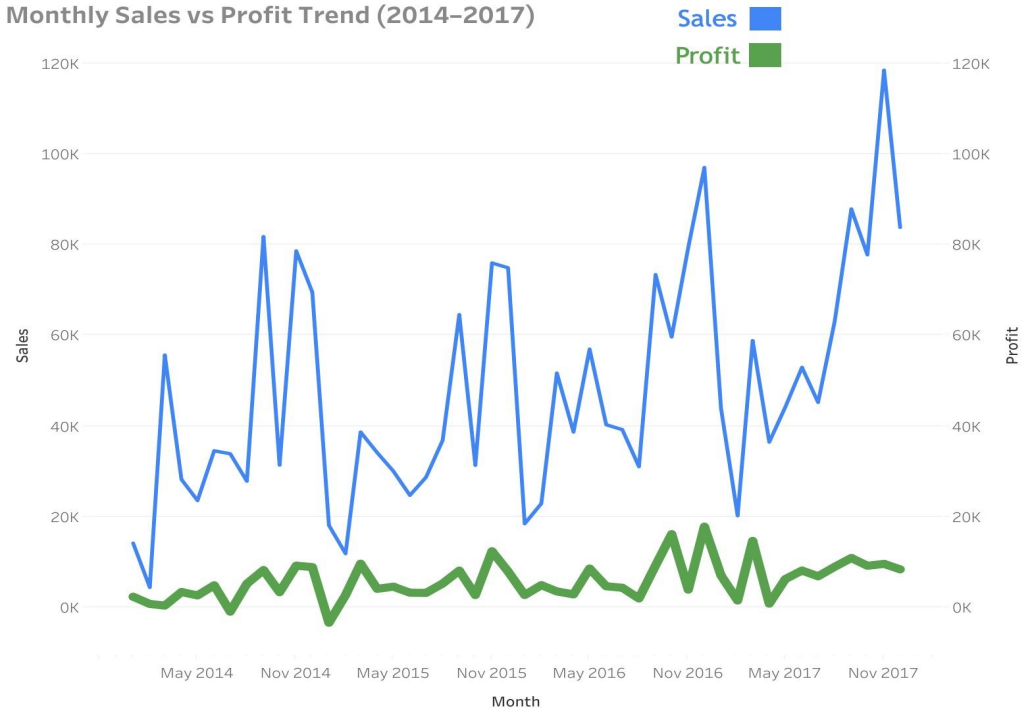
Key Business Questions

- How have **sales and profit trended** over time?
- Are **increases in sales** translating into higher **profitability**?
- How does **discounting** impact **profit margins**?
- Which product **categories** perform best in terms of **revenue** and **profit**?
- Are certain **categories** more reliant on **discounting** than others?

Sales vs Profit Trend Over Time

- Sales show a clear upward trend over time
- Profit is more volatile and does not consistently follow sales growth
- Periods of high sales do not always correspond to high profitability
- **This suggests underlying factors (such as discounting) impacting margins**

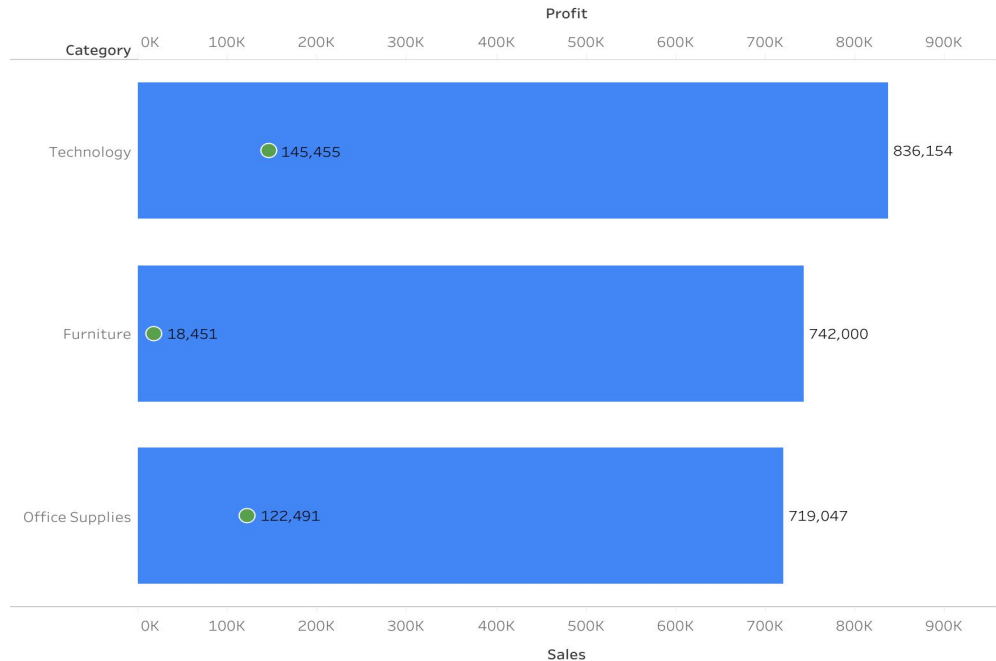
Monthly Sales vs Profit Trend (2014–2017)



Sales & Profit by Product Category

- Technology generates the highest sales and profit
- Office Supplies performs strongly in profit relative to its sales
- Furniture generates high sales but significantly lower profit
- **This suggests profitability varies across categories, not just revenue**

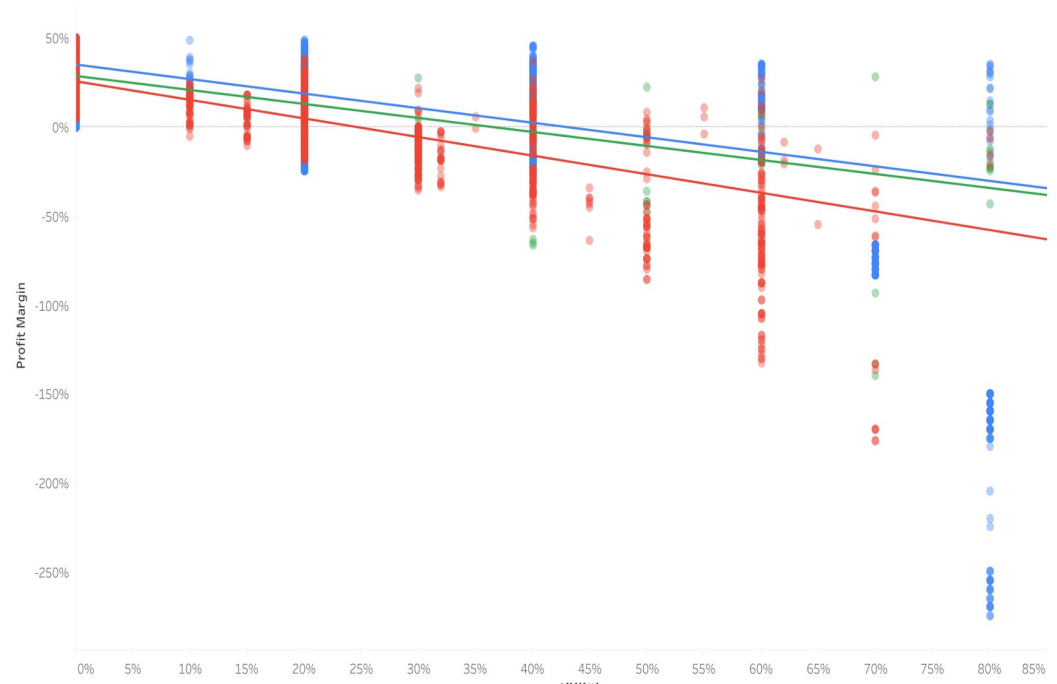
Sales & Profit by Product Category



Impact of Discounts on Profit Margins

- **There is a clear negative relationship between discounts and profit margin**
- As discount levels increase, profit margins consistently decrease
- High discount levels frequently result in negative profit margins
- Furniture shows the steepest decline, indicating higher sensitivity to discounting

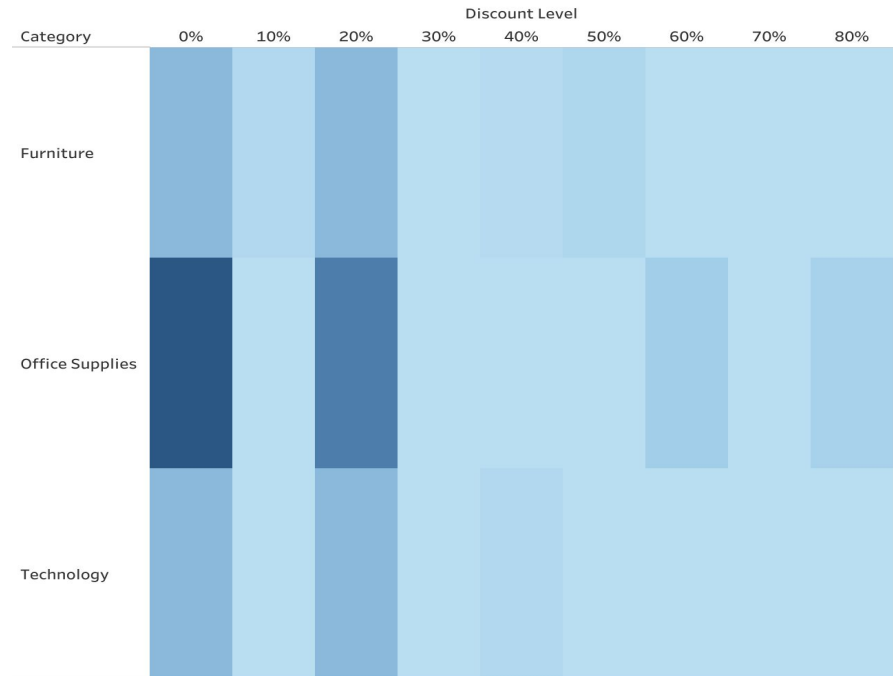
How Discounts Impact Profit Margins



Discount Usage by Product Category

- **Discounting is widely used across all product categories**
- **Office Supplies** shows the highest concentration of low-to-mid discounts
- Higher discount levels are used less frequently but still present across categories
- Discounting behaviour is consistent across categories, not isolated to one segment

Discount Usage by Product Category



Key Insights

- **Sales** have grown steadily over time, but **profitability** has not followed at the same rate
- Higher sales **do not always translate** into higher profit, indicating margin pressure
- **Discounting has a clear negative impact on profit margins across all categories**
- **Furniture** is the most affected category, showing the **steepest decline in profitability** with **increased discounting**
- Discounting is widely used across the business, suggesting a systemic reliance on price reductions

Recommendations

- **Reduce high discount levels**, particularly in categories with **low profit margins**
- Review pricing and cost structures in **Furniture** to improve profitability
- Focus on growing **high-performing categories** such as Technology
- Implement stricter **monitoring of profit margins** alongside sales performance
- Test alternative strategies (e.g. **targeted promotions**) instead of broad discounting

Appendix

Data Source

Superstore retail dataset (2014–2017)

Tools Used

- Python (data exploration & analysis)
- Tableau Public (data visualisation & dashboarding)

Key Notes

- Profit Margin calculated as $\text{Profit} \div \text{Sales}$
- Each row represents a single order
- Discount values represent % reductions applied per order

Project Links

- Tableau Dashboard:
<https://public.tableau.com/views/RetailE-CommercePerformance/RetailPerformanceOverview>
- GitHub Repository:
<https://github.com/tony-deehan/retail-performance-analysis>